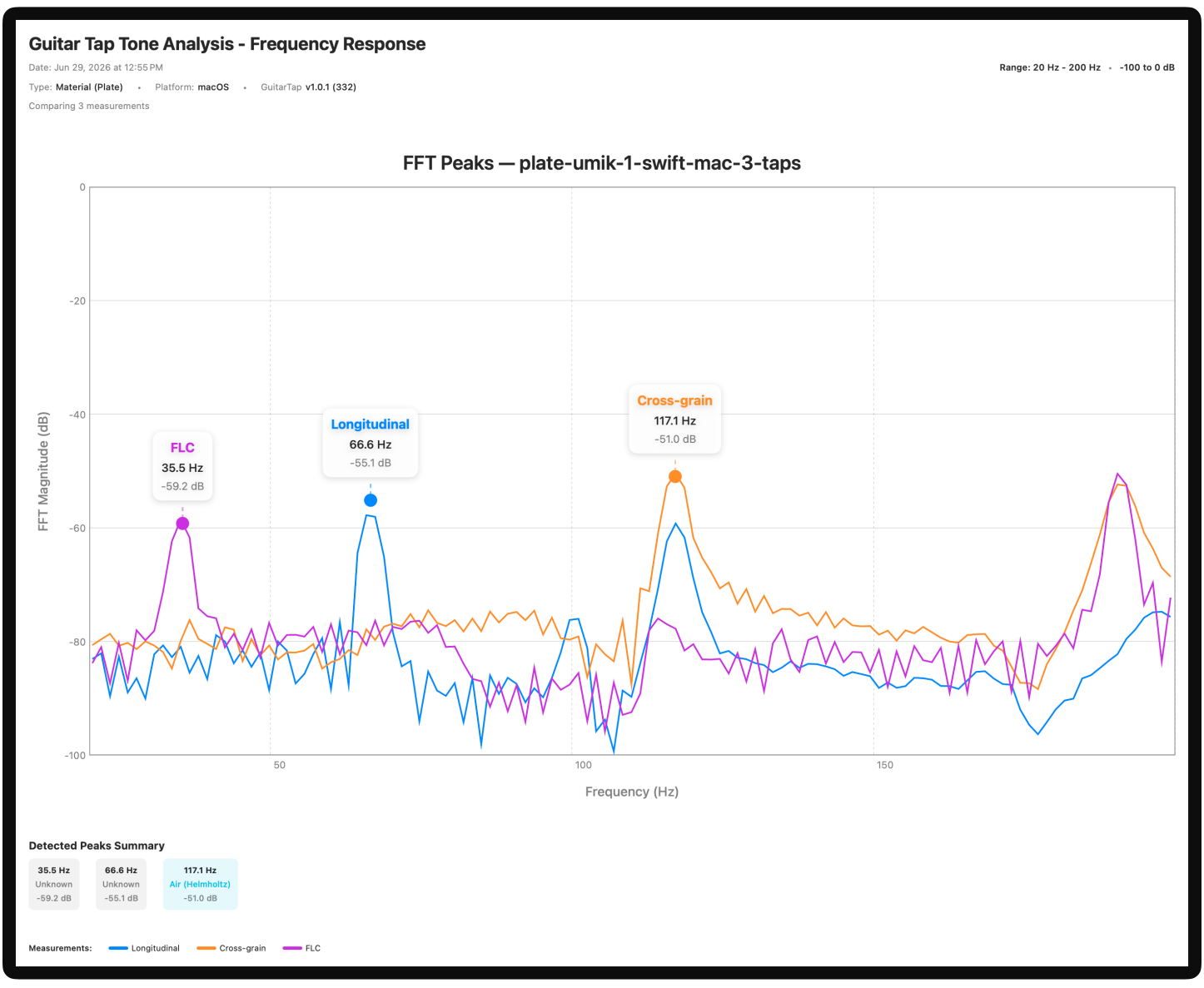
GuitarTap

Tap Tone Analysis Report

Jun 29, 2026 at 12:55 PM

Measurement Name:	plate-umik-1-swift-mac-3-taps
Type:	Material (Plate)
Notes:	3 taps per phase
Frequency Range:	20.0 Hz – 200.0 Hz
Microphone:	Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
35.5 Hz	-59.2 dB	C#1	8.0	FLC (Diagonal)
66.6 Hz	-55.1 dB	C2	15.0	Longitudinal (L)
117.1 Hz	-51.0 dB	A#2	20.0	Cross-grain (C)

Gore Target Thickness

2.89 mm

Body: 490 × 390 mm · f_vs = 60 (Classical Top)

GLC (Shear Modulus): 0.343 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**

Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **66.6 Hz** fC: **117.1 Hz** fLC: **35.5 Hz**

Speed of Sound (L): **4154 m/s**

Speed of Sound (C): **1142 m/s**

Young's Modulus (L): **6.02 GPa**

Young's Modulus (C): **0.46 GPa**

GLC (Shear Modulus): **0.343 GPa**

Specific Modulus (L): **17.3** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **11.9**

Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.076**
typical: 0.04 – 0.08

Long/Cross Ratio: **13.2**
typical: 12 – 25

Overall Quality: **Good**

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.